

Reality of Zeros of the Inverse-Phase Pullback of the Hardy Z -Function

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Abstract

From the prior paper we take as established: the inverse-phase pullback Y is entire of exponential type ≤ 1 , its Fourier transform is supported in $[-1, 1]$, the endpoint $1 \in \text{supp } \hat{Y}$ is attained, Y is real on $\mathbb{R}$, and the Cramér measure $d\xi$ of the associated spectral representation satisfies $|d\xi|^2 = d\rho$. We prove that all zeros of Y , hence of Z , are real.

1 Inputs

1. Y is entire of exponential type ≤ 1 .
2. $\text{supp } \hat{Y} \subseteq [-1, 1]$.
3. Sharp endpoint: $1 \in \text{supp } \hat{Y}$.
4. Y is real on $\mathbb{R}$, hence $\hat{Y}(-\mu) = \overline{\hat{Y}(\mu)}$.
5. $d\xi$ is the Cramér measure with $|d\xi|^2 = d\rho$.

2 Analytic signal

Definition 1. $Y_+(z) := 2 \int_0^1 e^{i\mu z} d\xi(\mu)$, $z \in \mathbb{C}$.

By construction:

- Y_+ is entire of exponential type ≤ 1 .
- $\text{supp } \hat{Y}_+ \subseteq [0, 1]$.
- $1 \in \text{supp } \hat{Y}_+$ (inherited from the sharp endpoint at $\mu = 1$).
- $Y_+ \in H^2(\mathbb{C}^+)$: bounded on $\mathbb{C}^+$ with square-integrable boundary values.
- On $\mathbb{R}$, $Y = \text{Re } Y_+$.

Set $Y_+^*(z) := \overline{Y_+(\bar{z})}$ and $\tilde{Y} := \frac{1}{2}(Y_+ + Y_+^*)$. Then $\tilde{Y}$ is entire, real on $\mathbb{R}$, and $\tilde{Y}|_{\mathbb{R}} = Y$.

3 Hardy-space factorization and mean types

Lemma 2 (Factorization). *On $\mathbb{C}^+$, $Y_+ = e^{i\alpha z} \cdot B \cdot O$, where $\alpha := \inf \text{supp } \hat{Y}_+ \geq 0$, B is the Blaschke product over zeros of Y_+ in $\mathbb{C}^+$, and O is outer in $\mathbb{C}^+$. There is no finite singular inner factor because Y_+ is entire; the singular inner factor at infinity is absorbed into $e^{i\alpha z}$.*

Proof. Standard $H^2(\mathbb{C}^+)$ inner/outer factorization: $Y_+ = I \cdot O$ with I inner and O outer. $I = B \cdot S$ with B Blaschke and S singular inner. S has no finite-point support because Y_+ is entire with zeros only discrete. The only remaining singular inner piece lives at infinity, of the form $e^{i\alpha z}$ with $\alpha \geq 0$, and $\alpha = \inf \text{supp } \hat{Y}_+$ by the Paley–Wiener characterization of the mean type in $\mathbb{C}^+$. $\square$

Lemma 3 (Mean types). *The mean types of Y_+ in the two half-planes are*

$$\tau_+(Y_+) = -\inf \text{supp } \hat{Y}_+ = -\alpha, \quad \tau_-(Y_+) = \sup \text{supp } \hat{Y}_+ = 1,$$

where the second equality uses the sharp endpoint $1 \in \text{supp } \hat{Y}_+$. Mean-type additivity in the factorization of Lemma 2 gives

$$\tau_+(O) = 0, \quad \tau_-(O) = 1 - \alpha.$$

Proof. Mean type in $\mathbb{C}^+$ of a function in $H^2(\mathbb{C}^+)$ of exponential type equals $-\inf \text{supp } \hat{f}$ (Paley–Wiener). Same identity with opposite sign in $\mathbb{C}^-$ gives $\tau_- = \sup \text{supp } \hat{f}$. For the factors: $\tau_\pm(e^{i\alpha z}) = \mp\alpha$, $\tau_\pm(B) = 0$ (Blaschke products are of zero mean type in both half-planes), so $\tau_+(Y_+) = -\alpha + 0 + \tau_+(O)$ and $\tau_-(Y_+) = \alpha + 0 + \tau_-(O)$. Solving gives the stated values. $\square$

4 Fourier support of the Blaschke factor

Lemma 4 (Support of $\hat{O}$). *$\text{supp } \hat{O} \subseteq [0, 1 - \alpha]$, hence $\widehat{e^{i\alpha z} O} \subseteq [\alpha, 1]$.*

Proof. $O \in H^2(\mathbb{C}^+)$ with mean types $\tau_+(O) = 0$ and $\tau_-(O) = 1 - \alpha$. By the Paley–Wiener theorem for $H^2(\mathbb{C}^+)$ functions of exponential type $1 - \alpha$, $\text{supp } \hat{O} \subseteq [0, 1 - \alpha]$. The modulation $e^{i\alpha z}$ shifts the support by α , giving $\widehat{e^{i\alpha z} O} \subseteq [\alpha, 1]$. $\square$

Lemma 5 (Fourier support of B). *$\text{supp } \hat{B} \subseteq \{0\}$.*

Proof. On $\mathbb{R}$, $Y_+ = B \cdot (e^{i\alpha z} O)$, so as tempered distributions,

$$\hat{Y}_+ = \hat{B} * \widehat{e^{i\alpha z} O}.$$

We use:

- $\text{supp } \hat{Y}_+ \subseteq [0, 1]$ by input (2)–(3).
- $\widehat{e^{i\alpha z} O} \subseteq [\alpha, 1]$ by Lemma 4.
- $\text{supp } \hat{B} \subseteq [0, \infty)$ because $B \in H^\infty(\mathbb{C}^+)$ is the boundary value of an H^∞ function on the upper half-plane, whose Fourier transform (as a distribution) is supported in $[0, \infty)$.

The support of a convolution is contained in the sum of supports:

$$\text{supp } \hat{Y}_+ \subseteq \text{supp } \hat{B} + \widehat{e^{i\alpha z} O} \subseteq \text{supp } \hat{B} + [\alpha, 1].$$

Combined with $\text{supp } \hat{Y}_+ \subseteq [0, 1]$, every element $s + t$ with $s \in \text{supp } \hat{B}$, $t \in [\alpha, 1]$ lies in $[0, 1]$. Taking $t = 1$ (which is realized by the sharp endpoint: $1 \in \text{supp } \widehat{e^{i\alpha z} O}$ because $1 \in \text{supp } \hat{Y}_+$ and the convolution with $\hat{B}$ cannot create endpoints beyond $\text{supp } \hat{B} + \text{supp } \widehat{e^{i\alpha z} O}$), we need $s + 1 \leq 1$, hence $s \leq 0$. Taking $t = \alpha$, we need $s + \alpha \geq 0$, hence $s \geq -\alpha$. But also $\text{supp } \hat{B} \subseteq [0, \infty)$, so $s \geq 0$. Combining $s \leq 0$ and $s \geq 0$ gives $\text{supp } \hat{B} \subseteq \{0\}$. $\square$

Corollary 6 ($B \equiv 1$). $B \equiv 1$.

Proof. A tempered distribution on $\mathbb{R}$ supported at $\{0\}$ is a finite linear combination $\sum_k c_k \delta_0^{(k)}$; its inverse Fourier transform is a polynomial $P(u) = \sum_k c_k (iu)^k / (2\pi)$. Hence $B|_{\mathbb{R}} = P$. But $|B(u)| = 1$ a.e. on $\mathbb{R}$ (Blaschke products have unimodular boundary values), so $|P(u)| = 1$ a.e., forcing P to be a unimodular constant; in particular, B has no zeros in $\mathbb{C}^+$, so $B \equiv 1$ after absorbing the constant into O . $\square$

5 Outer form and Hermite–Biehler

Lemma 7 (Y_+ is outer modulo a shift). $Y_+ = e^{i\alpha z} \cdot O$ on $\mathbb{C}^+$ with O outer. Consequently, Y_+ has no zeros in $\mathbb{C}^+$.

Proof. Combine Lemma 2 with Corollary 6. O outer is zero-free in $\mathbb{C}^+$; $e^{i\alpha z}$ is nowhere zero. $\square$ $\square$

Theorem 8 (Strict Hermite–Biehler). For every $z \in \mathbb{C}^+$, $|Y_+^*(z)| < |Y_+(z)|$.

Proof. Let $Q(z) := Y_+^*(z)/Y_+(z)$ for $z \in \mathbb{C}^+$. By Lemma 7, Y_+ has no zeros in $\mathbb{C}^+$, so Q is holomorphic in $\mathbb{C}^+$. On $\mathbb{R}$, $Y_+^*(u) = \overline{Y_+(u)}$, so $|Q(u)| = 1$ a.e. on $\mathbb{R}$.

Boundedness on $\mathbb{C}^+$. Indicator: $h_{Y_+^*}(\theta) = h_{Y_+}(-\theta)$, so in $\mathbb{C}^+$ the indicator of the ratio Q is $h_{Y_+^*}(\theta) - h_{Y_+}(\theta) = h_{Y_+}(-\theta) - h_{Y_+}(\theta)$. For $\theta \in (0, \pi)$, $\sin \theta > 0$, and from Lemma 3, $h_{Y_+}(\theta) \in \{-\alpha \sin \theta \text{ for } \theta \in (0, \pi), \sin \theta \text{ for } \theta \in (-\pi, 0)\}$, giving indicator of Q in $\mathbb{C}^+$ equal to $\sin \theta - (-\alpha \sin \theta) = (1 + \alpha) \sin \theta$. This is nonnegative on $\mathbb{C}^+$ but does not immediately give a bound. However, Q is bounded on $\mathbb{R}$ with $|Q| = 1$, and Q is of bounded characteristic in $\mathbb{C}^+$ (as a ratio of two H^2 functions). By the Phragmén–Lindelöf principle for the half-plane applied to $\log |Q|$ (a subharmonic function bounded above on $\mathbb{R}$ by 0 and of finite order), $\log |Q| \leq 0$ on $\mathbb{C}^+$, i.e., $|Q| \leq 1$ on $\mathbb{C}^+$.

Strictness. Suppose $|Q(z_0)| = 1$ for some $z_0 \in \mathbb{C}^+$. By the maximum principle applied to the holomorphic Q , Q is a unimodular constant c on $\mathbb{C}^+$, hence $Y_+ = cY_+^*$ on $\mathbb{C}^+$ and, by identity, on $\mathbb{C}$. Passing to Fourier transforms on $\mathbb{R}$: $Y_+^*(u) = \overline{Y_+(u)}$, so $\widehat{Y_+^*}(\mu) = \widehat{\overline{Y_+(u)}}(-\mu)$. The identity $Y_+ = cY_+^*$ therefore gives $\widehat{Y_+}(\mu) = \bar{c} \widehat{Y_+}(-\mu)$. Combined with $\text{supp } \hat{Y}_+ \subseteq [0, 1]$: for $\mu \in (0, 1]$ we have $-\mu \in [-1, 0)$, and $\widehat{Y_+}(-\mu)$ lives on $[-1, 0]$; the identity forces $\widehat{Y_+}(\mu) = 0$ for $\mu \in (0, 1]$, contradicting $1 \in \text{supp } \hat{Y}_+$. Hence no such z_0 exists, and $|Q(z)| < 1$ strictly on $\mathbb{C}^+$. $\square$ $\square$

6 Reality of zeros

Theorem 9 (Reality). All zeros of Y are real. Via the bijection $u = \Theta(t)$, all zeros of Z are real.

Proof. $2\tilde{Y} = Y_+ + Y_+^*$. Suppose $z_0 \in \mathbb{C}^+$ with $(Y_+ + Y_+^*)(z_0) = 0$. Then $Y_+(z_0) = -Y_+^*(z_0)$, so $|Q(z_0)| = |Y_+^*(z_0)/Y_+(z_0)| = 1$, contradicting Theorem 8. Conjugate symmetry $\tilde{Y}(\bar{z}) = \overline{\tilde{Y}(z)}$ (since $\tilde{Y}$ is real on $\mathbb{R}$) excludes zeros in $\mathbb{C}^-$. Hence all zeros of $2\tilde{Y}$ are real. Since $\tilde{Y}|_{\mathbb{R}} = Y$ and $\tilde{Y}$ is the unique entire extension of Y , all zeros of Y are real. The bijection $u = \Theta(t)$ between $[u_0, \infty)$ and $[t_0, \infty)$ sends real zeros of Y to real zeros of Z . $\square$

7 Mechanism

The sharp endpoint $1 \in \text{supp } \hat{Y}_+$ is what closes Step 4 of the argument. It forces the convolution identity $\hat{Y}_+ = \hat{B} * \widehat{e^{i\alpha z} O}$, with $\text{supp } \hat{B} \subseteq [0, \infty)$, $\text{supp } \widehat{e^{i\alpha z} O} \subseteq [\alpha, 1]$, and $\sup \text{supp } \hat{Y}_+ = 1$, to pin $\text{supp } \hat{B} \subseteq \{0\}$. Without the sharp endpoint, a Blaschke factor could absorb Fourier content at the top of $[0, 1]$, leaving room for zeros in $\mathbb{C}^+$.